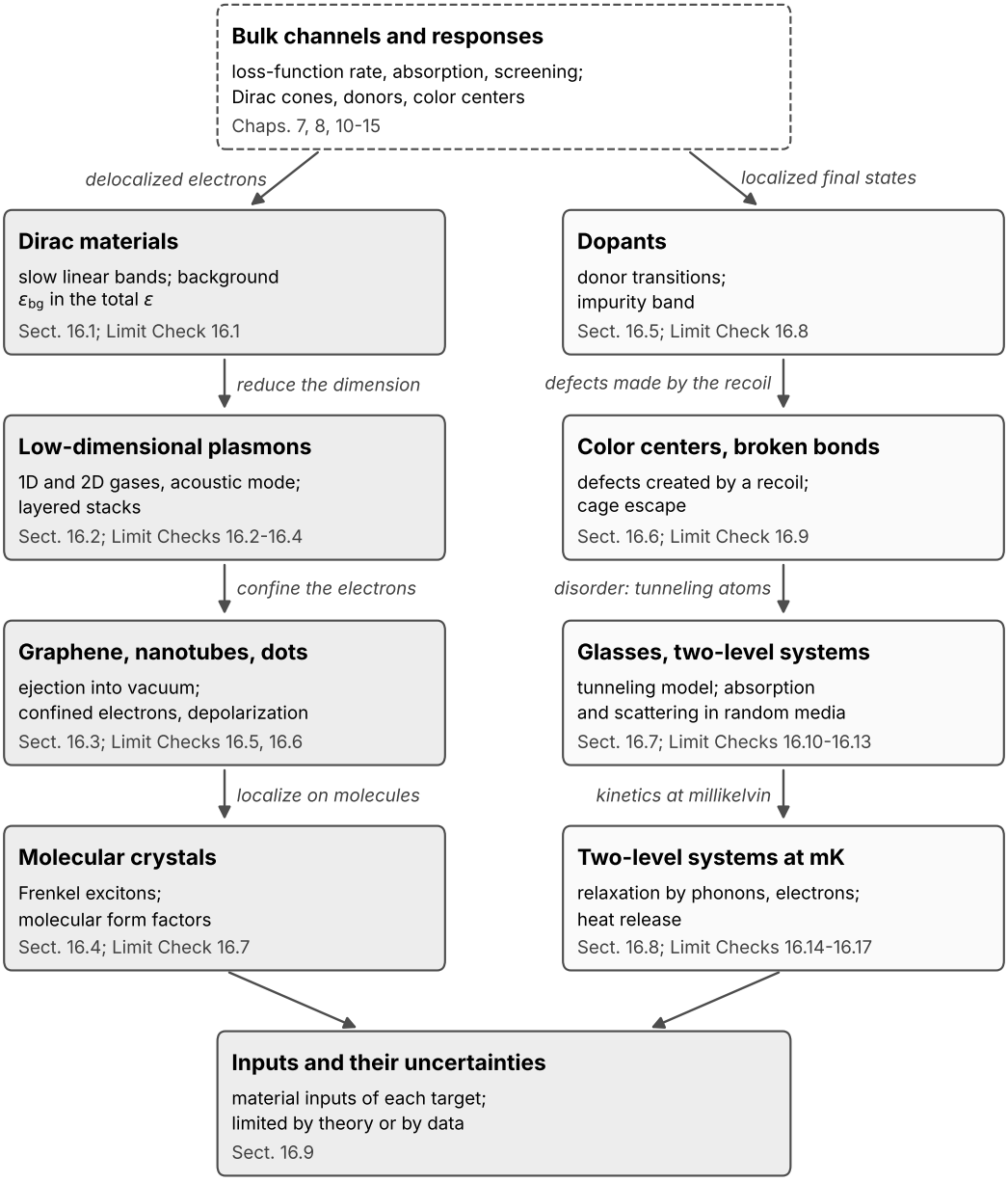
Bulk channels and responses
loss-function rate, absorption, screening;
Dirac cones, donors, color centers
Chaps. 7, 8, 10-15
delocalized electrons
localized final states
Dirac materials
slow linear bands; background
ε_{bg} in the total ε
Sect. 16.1; Limit Check 16.1
Dopants
donor transitions;
impurity band
Sect. 16.5; Limit Check 16.8
reduce the dimension
defects made by the recoil
Low-dimensional plasmons
1D and 2D gases, acoustic mode;
layered stacks
Sect. 16.2; Limit Checks 16.2-16.4
Color centers, broken bonds
defects created by a recoil;
cage escape
Sect. 16.6; Limit Check 16.9
confine the electrons
disorder: tunneling atoms
Graphene, nanotubes, dots
ejection into vacuum;
confined electrons, depolarization
Sect. 16.3; Limit Checks 16.5, 16.6
Glasses, two-level systems
tunneling model; absorption
and scattering in random media
Sect. 16.7; Limit Checks 16.10-16.13
localize on molecules
kinetics at millikelvin
Molecular crystals
Frenkel excitons;
molecular form factors
Sect. 16.4; Limit Check 16.7
Two-level systems at mK
relaxation by phonons, electrons;
heat release
Sect. 16.8; Limit Checks 16.14-16.17
Inputs and their uncertainties
material inputs of each target;
limited by theory or by data
Sect. 16.9